

PREDICTING THE OUTCOME OF SUMO WRESTLING BOUTS

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Abstract

The goal of this research was to create a model that could predict the outcome of sumo wrestling matches. An overview of the many machine learning models or strategies utilised to make this prediction. Sumo wrestlers' wins and losses have been analysed and machine learning models were trained using historical data from 2010 to 2023. Three machine learning techniques are employed in this paper: logistic regression, K nearest neighbors, Decision Tree and Random Forest. According to the results, the Logistic Regression train accuracy was 59.52%, while the test accuracy was 58.65%; the Decision Tree train accuracy was 55.08%, with a test accuracy of 53.69%; the k nearest neighbors train accuracy was 65.41%, with the test accuracy 55.75% and the Random Forest Train accuracy was 57.76%, with a test accuracy of 56.45%.

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Chapter 1

Introduction

This chapter gives an insight into my dissertation, which delves into the domain of forecasting sumo wrestling outcomes. It travels through Sumo's historical environment, merging tradition with modern data-driven investigation. The fascination with uncovering hidden patterns throughout the sport's past drives this research. This study sets out on a multidimensional path with the goal of forecasting sumo match results utilising historical data and sophisticated machine learning techniques. The extensive information includes not only wrestlers' prior performance records and rankings but also age, height, and weight, enhancing the prediction endeavour. This ambition connects the past and the future by combining Sumo's tradition with the analytical capabilities of current approaches. This study uses machine learning methods to reveal the opaque terrain of sumo wrestling predictions, adding a new dimension to both sports tradition and data-driven discovery.

1.1 Background and Motivation

Sumo wrestling, an ancient Japanese sport, has a millennium-long history. In this extended span, it was deeply woven into the fabric of Japanese society, playing an important role in a variety of rituals, festivities, and entertainment acts. The sport's historical footprint is not limited to anecdotal narratives, but is supported by physical artefacts of clay and earthenware sculptures that clearly depict events from early sumo matches. These relics serve as invaluable historical artefacts, demonstrating unequivocally that sumo's ubiquity and significance reached deep into Japan's ancient origins. These historical gems do more than simply prove sumo's existence; they also

provide a window into the sport's progression throughout time. We see the transforming path of sumo wrestling through the prism of these artefacts, uncovering the customs, traditions, and rituals that accompanied its inception. Beyond the physical prowess demonstrated in the ring, these relics reveal the cultural fabric into which the sport was tightly woven (Ochiai & Takeda 2001). Furthermore, these artefacts provide insight into sumo's significant societal effect. Their presence attests not just to the sport's athletic prowess, but also to its pivotal role in defining Japanese society's collective identity. These historical artefacts continue to resound the long legacy of sumo wrestling, repeating its cultural and social significance across generations, in a period where traditions develop and heritage fuses with the current (Duggan & Levitt 2002) .

According to Ochiai & Takeda (2001), sumo wrestling's features changed dramatically between 710 and 794, during the late Nara era. The sport's roots, formerly interwoven with Shinto shrines—hallowed places of worship in Japan—began to merge with a new narrative. It rose to prominence as a main attraction during the imperial court's yearly festivals, transitioning from a religious affiliation to a focal point of cultural entertainment. Sumo made another revolutionary step during the Kamakura era, which lasted from 1185 until 1392. It transitioned from its previous duties to that of a martial art, having an important place in the physical training routine of the prestigious samurai class. Sumo, with martial prowess at its core, went beyond entertainment to become a cornerstone of training soldiers for the difficulties of their day. Sumo's voyage intersected with the Edo era, which lasted from 1615 until 1867. It was known as the Tokugawa period, and it marked the beginning of a new chapter in the history of the sport. Sumo, no longer a mere ritual or practise, took front stage, grabbing the hearts of the expanding middle class as a public spectacle. During this time, it was transformed into a beloved national sport, with a popularity that resonated across socioeconomic strata. Sumo exhibited its varied nature by building this historical tapestry. Sumo's journey reflects the progress of not just a sport but a monument to the evolution of Japan's societal fabric, from religious ceremonies to physical preparation for warriors to a countrywide cultural phenomenon.

1.1.1 Sumo Bouts

The arena of sumo wrestling is a sacred circle known as the dohyo. This time-honored tradition has its origins intertwined with the past, dating back to the Edo period, an era spanning from 1615 to 1867. During this time, the dohyo's dimensions were confined within a diameter of 394 cm. Yet, the dynamism of the sport and the quest for added thrill propelled its transformation. With a nod to accommodating bulkier wrestlers and elevating the excitement factor, the dohyo's dimensions expanded to a striking 455 cm. This circular stage holds more than just dimensions—it cradles the soul of sumo wrestling. The wrestlers, propelled by tradition and honor, tread upon its hallowed ground to demonstrate their prowess and mettle. The dohyo's building is a laborious process rich in ceremonial importance. It is said to be the home of the Shinto deity of sumo, providing a spiritual dimension to the competition. The dohyo has been an unshakable symbol of Sumo's culture for years. It sits like a sentinel, repeating the rhythms of bygone rites and symbolising the bloodline of tradition Ochiai & Takeda (2001).



Figure 1.1: The muscle training by pushing the opponent across the sumo ring, dohyo.

Even in our modern era, the dohyo retains its place of dignity, a monument to the sport's rich cultural links. Each tournament sees the laborious preparation of this treasured venue, which is meticulously kept as a living testament to Sumo's long tradition. The expanded proportions of the dohyo throughout the Edo period are more than a mere change; they are a representation of Sumo's growth, a visible testament to its enduring appeal as an enduring Japanese legacy Ochiai & Takeda (2001).

1.1.2 Rules of Tournament

According to West (1997) sumo, despite its long history, is far from a static sport. Officially, there are 70 distinct rules that define the path to victory. The technique of pushing, throwing, thrusting, and driving one's opponent beyond the confines of the ring, or dohyo, lies at the heart of winning. As the story continues, some activities such as clinched finger grasping, hair pulling, abdomen kicking, and finger twisting are officially forbidden. The dohyo, identifiable by its distinctive white lines, functions as a key stage. Its boundaries govern not just the scope of activity, but also the boundaries of interaction. Any contact by a player or opponent with the sacred ground is deemed defeat. These precisely crafted standards respect the ideals of justice and provide a level playing field. Wrestlers must plot, execute, and exhibit their strength inside these sacred lines, all while keeping their own feet anchored and their opponents on their toes. The official referee, a sentinel within the dohyo, is an important person in this story. The referee orchestrates the flow of the play with watchful eyes, guaranteeing conformity to the rules and arbitrating each action. Meanwhile, the judges, stationed beyond the dohyo's limits, play an important role. Their perceptive assessment is crucial in deciding the final victor. Sumo develops as a symphonic symphony of strategy and talent in this dance of tradition and competitiveness. The dohyo remains the focal point of this spectacle, a stage of tradition and rigour where every move and moment defines the essence of this age-old struggle.

Sumo, unlike Western wrestling or boxing, does not use weight divisions, allowing larger rikishi to have an edge. However, age is an important factor in joining this sector. Males must be over the age of 23 and stand at a minimum height of 173 cm to become a sumo apprentice. Weight, which is an important factor in this activity, should be greater than 75 kg. Notably, a basic knowledge of the law is also essential. This mix of age, physicality, and education reveals the sumo journey's diverse demands, where heritage meets modernity and knowledge coexists with power (Kanehisa et al. 1997).

1.1.3 Ranking System of Sumo

Ochiai & Takeda (2001) and West (1997) says, the Japan Sumo Association matches include 700 to 900 professional sumo wrestlers. Wrestlers at the top level compete in a

daily match during a tournament. Lower-level wrestlers, on the other hand, compete in seven bouts. The banzuke, which acts as the professional sumo ranking list, is divided into six categories. The top level, Makuuchi, is made up of the best wrestlers. This division is further subdivided into five ranks: yokozuna, ozeki, sekiwake, komusubi, and maegashira. Juryo is the second-highest level, which houses wrestlers on the verge of entering the top division or those who have been relegated from Makuuchi.

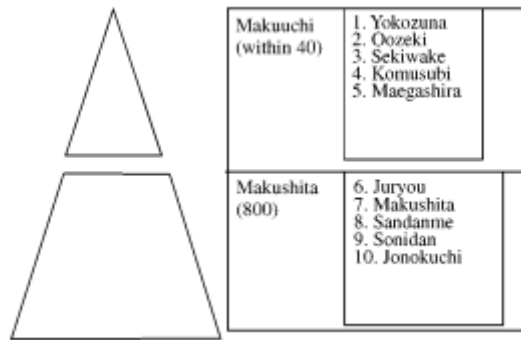


Figure 1.2: Sumo Ranking Hierarchy.

The Juryo category also has its own set of ranks. Makushita is the third division, which houses rising wrestlers. Sandanme, the fourth level, is for wrestlers who want to improve their talents and acquire experience. The fifth division, Jonidan, is for younger wrestlers improving their sumo skills. Jonokuchi, the sixth and lowest tier, is for recruits and those yet to make their professional debut. The banzuke is changed before to each sumo tournament to reflect recent performances and promotions or demotions among the wrestlers. It is an important instrument for choosing match pairings in the next competition, reflecting Sumo's competitive spirit. The Makuuchi division has particular rankings in declining order, such as Yokozuna, the apex rank for grand champions, which is generally held by just a few wrestlers. Ozeki: The second-highest rank attained by top-performing wrestlers. Sekiwake: The third-highest rank for individuals who excel slightly below the ozeki level. Komusubi: The fourth-highest level, given to potential wrestlers on the verge of higher ranks. Maegashira: The fifth-highest rank, covering wrestlers of varying experience levels and performances. It is the highest level in the Makuuchi division West (1997).

1.2 Aim and Objectives

Sumo wrestling, a beloved traditional sport deeply rooted in Japanese culture, has garnered widespread attention in Japan as well as a passionate global fan base. Nonetheless, sumo has not received the same amount of scholarly scrutiny as other sports. Because sumo is deeply ingrained in Japanese culture, much of the present study has been conducted by Japanese scholars, potentially limiting access to sumo-related studies for non-Japanese researchers due to language barriers. Furthermore, the availability and consistency of past sumo data may pose challenges, prohibiting researchers from conducting thorough analyses. To close this gap, we created a machine learning model that can predict sumo wrestling outcomes. This novel technique aims to provide fresh insights into the dynamics of the sport and maybe find patterns that contribute to match outcomes. Three different machine learning techniques are used in this paper: decision tree, logistic regression, K closest neighbours, and random forest. To put it simply, training a classification model with a dataset to predict specified categories or classes. This model gets excellent at guessing the category of fresh, previously unknown data after being trained. The goal of classification is to predict the target variable value, which represents the category or class that we want to identify. We hope to bring a new viewpoint on sumo wrestling through this attempt by leveraging the potential of machine learning and data analysis.

The associated objectives are:

- 1) Collecting sumo wrestling history data from previous years to the present.
- 2) An in-depth analysis of current material is being studied in order to obtain insights into the complexities of the sumo wrestling ecology, player categorizations, and the extensive rule structure that governs the sport.
- 3) Dive into a complete grasp of the multiple features of sumo wrestling, such as the complexities of the ranking system, the relevance of height and weight, wrestlers' demanding training regimes, and the complicated dynamics of match systems.
- 4) The removal of unneeded and redundant elements can also improve model performance. This method demands a good understanding of topic expertise, as well

as familiarity with technical vocabulary and sumo wrestling complexities.

- 5) Choosing the most accurate machine learning algorithm to evaluate performance.

1.3 Research Project Work Plan

The major goal of this project is to create a machine learning model capable of reliably predicting the outcomes of sumo wrestling contests, a complicated and culturally significant sport. This endeavour is a multi-step procedure that begins with the painstaking collection of historical data on sumo wrestling, including specific information about the wrestlers or rikishi, such as their ranks, physical attributes, and detailed prior performance records. The collection of data pertaining to the characteristics of each bout, such as the tournament location, the specific type of bout (ranging from yokozuna vs. ozeki clashes to other variations), and the prevailing match conditions, is equally important, all of which contribute to the intricate dynamics of this ancient sport.

In the ensuing data processing step, the data is rigorously cleaned to avoid mistakes, and missing values are properly handled. Furthermore, outliers and data inconsistencies are recognised and addressed to maintain the dataset's integrity. Wrestler ranks, for example, are turned into numerical representations, making them susceptible to machine learning methods.

Feature selection and engineering emerge as critical activities in which the research team attempts to find the most relevant aspects influencing results. These might include a variety of characteristics such as wrestler rankings, historical win-loss ratios, physical qualities, and prior tournament performance records. To improve the model's predictive capacity, new elements might be added or current ones integrated.

Model selection is a vital step, with a variety of classification-focused machine learning methods such as random forests, logistic regression, support vector machines, decision trees and neural networks being considered. The best model is one that balances performance measures with interpretability, and is closely aligned with the special needs of forecasting sumo wrestling results.

Model training entails dividing data into training and testing sets to aid the learning process of the model. Cross-validation approaches, notably k-fold cross-validation, are

used to assure robustness and generality.

Model evaluation goes beyond accuracy to include a thorough examination of precision, recall, F1 score, and other relevant measures. Confusion matrices and ROC curves, for example, reveal subtle insights into the model's performance characteristics.

The repeated process of hyperparameter tuning refines the model further by maximising its prediction ability using strategies such as grid search or random search.

The interpretation of the model's outputs is critical, digging into the aspects that have the greatest effect on outcomes. Real-world validation is used to get insights into the model's strengths and weaknesses.

Finally, this study results in a thorough report containing findings and insights. The final machine learning model, which includes accuracy and assessment measures, is a powerful tool for forecasting sumo wrestling match results. Beyond its immediate use, the study investigates deeper implications and future applications in a variety of scenarios, providing insight into the interesting junction of tradition and technology in the field of sports analytics.

1.4 Project Overview

The first chapter introduces the world of sumo wrestling, providing a historical framework for understanding its origins and cultural significance. This lays the groundwork for the subsequent in-depth research, which delves into several areas of sumo, such as its regulations, ranking structure, win-loss dynamics, and finer nuances. This substantial research serves as the foundation for the paper's content.

The second chapter's literature review offers insight on the machine learning algorithms that serve as the cornerstone of the project's methodology. It not only sheds light on these algorithms, but it also stresses the crucial importance of significant scholarly articles that have affected the project's trajectory. These papers provide a significant contribution to the correct forecast of sumo matches.

The third chapter goes into the dataset's analysis, design, and representation in further detail. This section goes beyond surface-level knowledge to provide critical insights into the complex network of machine learning algorithms used to create a

prediction model with exceptional accuracy for sumo matches.

In the fourth chapter, the implementation phase takes centre stage, as theoretical concepts from previous portions are transformed into actual, tangible operations. This chapter painstakingly describes the process of converting theoretical knowledge into practical practise.

Review Results and Discussion, the fifth chapter, is critical in disclosing the outcomes of a multidimensional review process. The discussion that follows focuses on the performance of several machine learning models, with the objective of selecting the best model for predicting accurate sumo matches.

The study culminates in the sixth chapter with a thorough explanation of the machine learning techniques employed throughout. Furthermore, this chapter enables broad comparisons based on a wide range of evaluation indicators, resulting in a comprehensive understanding of the research's achievements.

1.5 Summary

The project intends to overcome the gap in academic evaluation of sumo wrestling by applying machine learning algorithms to predict match results. While sumo has a committed global following, academic study on the sport has been limited by language barriers and data access concerns. To solve this, the project employs machine learning techniques such as decision trees, logistic regression, K closest neighbours (KNN), and random forests. These algorithms are trained on historical sumo wrestling data in order to detect trends and provide insight into the complex dynamics that influence match outcomes. The project is divided in stages, beginning with data collecting and finishing with model construction.

The following tasks are on the agenda: data preparation, feature selection, model training, assessment, hyperparameter modification, and interpretation. The research aims to provide fresh insights into this ancient sport by thoroughly assessing current information and delving into topics such as sumo wrestling's ranking system, physical characteristics, and training procedures. The effort intends to investigate the intricacies of sumo wrestling while overcoming linguistic and data-related barriers via the use of machine learning and data analysis. Finally, this research study tackles the problem

of applying machine learning to predict sumo wrestling outcomes, with the objective of overcoming linguistic limitations and data shortages. The project's purpose is to use numerous machine learning approaches and a specified work plan to uncover new insights and contribute to a fuller understanding of the dynamics behind sumo wrestling contests.

Chapter 2

Literature Review

In this essay, sumo wrestling is briefly discussed along with its historical background, cultural relevance in Japan, and appeal as a traditional sport. The literature on sports outcome prediction utilising machine learning techniques (such as random forest, logistic regression, decision trees and k nearest neighbors) is examined in this paper. Additionally, the goal of this work is to examine earlier studies that used sumo wrestling data for a variety of analysis, including wrestler performance measures, ranking systems, and tournament statistics. The existing literature has to be critically examined and synthesised, identifying the gaps and contributions provided by earlier studies. The literature study will serve as the basis for your research paper and show how well you are versed in the field's body of knowledge. A thorough examination of current scholarly works, publications, and pertinent sources regarding sumo wrestling, sports prediction, and machine learning would be included in the literature review for a research paper on forecasting the results of sumo wrestling matches. The literature review should address the following important topics:

2.1 Related Work

Sánchez-Ruiz (2015) has performed a rigorous analysis to anticipate the fighting outcome between the army groups. The purpose is to take a close look at the accuracy of the prediction during the combat or battle. The classification techniques used in this study experiment are LDA (Linear Discriminant Analysis), kNN (k-Nearest Neighbour), SVM (support vector machines), QDA (Quadratic Discriminant Analysis) and

kkNN (Weighted K-Nearest Neighbour). The classes were separated using a linear discriminant analysis classification technique. The quadratic discriminant analysis (QDA) approach is used to determine if the variability or dispersion of data among distinct classes of the target variable (groups or categories) is the same. Non-linear classification is performed using support vector machines. The classifiers were assessed based on their setup settings and accuracy, which was calculated as the proportion of properly categorised samples. To discover the optimal parameters for each classifier, 10-fold repeated cross-validation was performed, averaging accuracy across 32 executions with an 80% training and 20% test dataset split. The simple classifier simply predicts the winner based on the percentage of samples in each class. quadratic discriminant analysis (QDA), Linear discriminant analysis (LDA) and support vector machines (SVM) achieved accuracy values ranging from 72% to 76%. The two instance-based approaches, on the other hand, demonstrated superior accuracy. Weighted K-Nearest Neighbour (KNN) predicted the winner with an astonishing 95.70% accuracy, implying that local-based methods perform better in this scenario. LDA, QDA, and SVM achieved 90% accuracy by the fourth quarter of the game, but KNN maintained 66% accuracy from the start, accurately categorising 90% of game states. The research might be developed by investigating more potential combat techniques as the volume and diversity of units rise, resulting in more complicated problems.

Prasetio & Harlili (2016), from the department of informatics engineering at Institut Teknologi Bandung examined using a logistic regression model to predict football match results. The study concentrated on forecasting the outcomes of matches in the 2015/2016 Barclays Premier League season, namely whether they would be won at home or away. Unlike some prior efforts that depended on guessing to identify relevant factors, this study relied on variables deemed significant by other researchers in the area. Furthermore, data from the FIFA video game was included in the prediction process to improve the model's accuracy. The logistic regression model was developed utilising data from the 2010–2011 season through the 2015–2016 season. The model's important variables were "home offence," "home defence," "away offence," and "away defence." The studies involved changing the seasons of the training data that was used to assess the model's prediction accuracy. The constructed model's prediction accuracy

was reported to be 69.5 percent. This indicates that in trials utilising multiple sets of training data, the model accurately predicted the results of around 69.5 percent of the testing data matches. This degree of precision shows how well the model's predictions matched the actual match results. According to the study, the accuracy improvement was remarkable when compared to earlier efforts in the same domain.

Overall, the study validated the logistic regression classifier's usefulness in predicting football match results and highlighted its potential to give significant insights for enhancing prediction accuracy in the sports sector.

The research paper titled "Prediction of the outcome of a Twenty-20 Cricket Match" by (Shenoy et al. 2023) gives an in-depth examination of the use of machine learning algorithms to forecast the outcomes of Twenty-20 (T20) cricket matches. The research investigates three alternative techniques for forecasting match outcomes and assesses their performance using several machine learning classifiers. The first method involves using player performance information, ratings, and clustering approaches to generate player ratings. The second method includes an AdaBoost classifier combined with ratings collected from paired interactions between batters and bowlers. The third method use k-means clustering to provide batting and bowling ratings, followed by classification using candidate classifiers.

For prediction tasks, the article heavily relies on machine learning methods such as AdaBoost, Decision Trees, Naive Bayes, Random Forests, Support Vector Machines (SVMs), and Bagging. The authors investigate several permutations of these algorithms for predicting match outcomes based on player data, ratings, and grouping. They use approaches including cross-validation, temporal scaling, and neighbourhood-based rating changes to improve the prediction models' accuracy. The research entails preprocessing the dataset, collecting key features, and training classifiers to predict whether Team 1 or Team 2 would win a T20 cricket match.

The research paper provides accuracy percentages for different machine learning algorithms, as in feature-based ratings. Adaboost: 52.76%, Randomised Forest: 51.77%, Multinomial Naive Bayes: 51.15%, Multinomial Naive Bayes (Bagging): 51.65%, Decision Trees: 51.87%, Linear SVM: 50.87%, Decision Trees (Bagging): 53.4%, Linear SVM (Bagging): 50.24%, Non-Linear SVM: 50.59%, Non-Linear SVM (Bagging):

50.4%. In modelling Player Batting and Bowling Ratings, Randomised Forest: 62.70%, Adaboost: 57.74%, Multinomial Naive Bayes: 57.23%, Multinomial Naive Bayes (Bagging): 58.34%, Decision Trees: 60.89%, Decision Trees (Bagging): 60.36%, Linear SVM: 60.84%, Non-Linear SVM (Bagging): 63.89%, Linear SVM (Bagging): 61.77%, Non-Linear SVM: 63.05%, Ensemble of all the above: 64.62%

In research paper, Hucaljuk & Rakipović (2011) focuses on applying machine learning algorithms to forecast the outcomes of football matches, particularly those in the Champions League. The article tackles the complexities of predicting football match results owing to the multiple aspects that contribute to the ultimate result, such as team form, individual player ability, team rankings, injuries, and even psychological factors such as fan effect. The authors describe a software method designed to address this prediction difficulty. The algorithm is supposed to anticipate match outcomes with roughly 60% accuracy, outperforming a reference technique that most likely forecasts the most common outcome. The article describes the system's development process, which includes feature selection and classifier selection. The authors emphasise the necessity of selecting relevant variables that have a major influence on match results. Team form, past meeting outcomes, team rankings, injured players, and average goals scored and received each game are all factors considered. The researchers conduct experiments to discover the best collection of features and assess several classifiers, such as LogitBoost, Naive Bayes, k-nearest neighbours, random forest, Bayesian networks and artificial neural networks. The findings demonstrate that the artificial neural network obtains the maximum accuracy, up to 68%, whereas the other classifiers perform differently. The findings imply that combining the correct characteristics and classifiers can result in adequate forecasting capabilities for football match outcomes.

In essence, the study shows how machine learning algorithms may be used to forecast football match outcomes in the Champions League. The proposed software solution outperforms the given aim of 60% prediction accuracy, with the artificial neural network outperforming the other evaluated classifiers. The study underlines the necessity of feature selection and classifier selection in making accurate predictions, as well as the difficulties of modelling and forecasting complex events such as football matches.

2.2 Summary

The research articles featured here investigate the use of machine learning and statistical approaches to forecast outcomes in a variety of areas, including combat situations, football matches, cricket matches, and Champions League football. The accuracy of several classification and regression models is thoroughly examined in these research, illustrating the ability of data-driven techniques to anticipate results. Notable discoveries include the effectiveness of instance-based approaches in battle outcome prediction, logistic regression in football match prediction, and the usefulness of machine learning systems in forecasting cricket match results. Furthermore, the difficulties of modelling and forecasting complex events, such as football matches, highlight the necessity of feature selection and classifier selection. These research efforts contribute to the improvement of predictive modelling in a variety of disciplines.

Chapter 3

Methodology Design

This chapter discusses our technique for examining the results of sumo wrestling matches. We'll go over the design and analytic methodology we used, providing light on the different measurements and assessment methodologies adapted particularly to this enthralling sport. Our goal is to give a thorough knowledge of the elements that influence the outcomes of sumo wrestling contests. To do this, we have designed a technique that takes into consideration sumo's unique dynamics and traditions. Throughout this chapter, we'll go through the processes we used to gather and analyse data to ensure the accuracy and relevance of our results. We'll also look at the statistical tools and analytical procedures that have been used for a thorough examination of the complex world of sumo wrestling results.

Furthermore, we'll provide a variety of metrics designed expressly to evaluate sumo wrestlers' performance and strategy in the ring. These metrics will help us to obtain a better understanding of what differentiates triumph from failure in this old and renowned sport. Finally, we'll go over the assessment techniques we've put in place to critically evaluate our study findings. These methodologies will serve as a vital checkpoint, verifying the validity and importance of our findings in the context of sumo wrestling.

3.1 Design of Methodology

We found a plethora of distinct cases in our dataset, each capturing the spirit of a sumo match. Every piece of data we have at our disposal, whether numerical or categorical, adds to the complex tapestry of this research.

This article sets out to present a novel approach for forecasting sumo wrestling bout results. We use historical data to train our dataset using a variety of machine learning classifiers. These classifiers, each with its own technique, will be scrutinised in order to find the one with the highest precision and prediction accuracy.

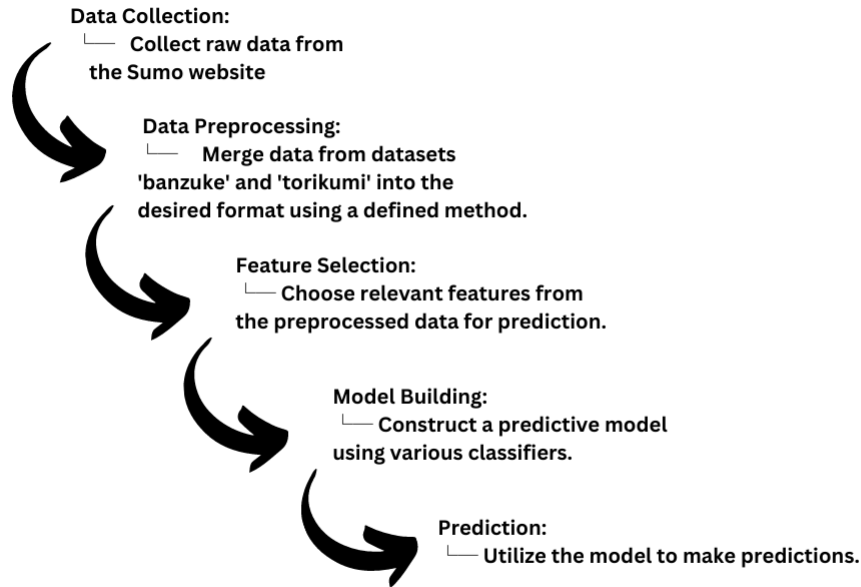


Figure 3.1: Methodology design framework of predicting sumo wrestling game

In the field of prediction, we have a broad variety of categorization algorithms at our disposal. From the cryptic Random Forest to the intuitive Decision Tree, and from the calculating Logistic Regression to the complex K-Nearest Neighbours, each algorithm brings a new viewpoint to the effort to predict the outcomes of these tense sumo battles. We hope to uncover the algorithm that stands out as the beacon of accuracy in this convoluted area of sport prediction through rigorous comparison.

Figure 3.1 demonstrates, in the first stage (data collection) we have collected raw data from sumo website. In the second stage we have merged the two data set "banzuke" and "torikumi", in third stage we have feature selection process where we have selected the relevant features from the processed data. Model building is the sixth stage where we construct various classifiers and lastly checked the prediction by utilizing the models

3.2 Analysis of the methods

Machine learning comprises a variety of approaches that range in their approach, the nature of the data they manage, and the challenges they solve. It is roughly classified into three types: supervised learning, unsupervised learning, and reinforcement learning. Within the area of machine learning, each kind serves a specific purpose.

In this research, we employed supervised learning, which includes the development of prediction models using both input and output data. This method is especially effective when the algorithm has to learn from labelled instances in order to generate good predictions on fresh, unseen data. It serves as the foundation for tasks such as classification and regression, in which the algorithm learns to map inputs to certain outputs.

3.2.1 Logistic Regression

Logistic regression is an extension of linear regression that is meant to handle situations where the dependent variable is either binary or multi-class with discrete answers. When dealing with binary outcomes, logistic regression applies sigmoid functions within a statistical framework. This mathematical technique allows for the examination of binary classification issues by translating real values into a range between 0 and 1, which corresponds to the probability distribution. Logistic regression, like linear regression, uses predefined coefficients to calculate the output for a given input. The ability of logistic regression to provide binary outcomes of 0 or 1 distinguishes it from linear regression models. This methodological distinction provides a valuable tool for solving numerous prediction and classification challenges in a variety of domains.

Logistic regression is used to categorise low-dimensional data with nonlinear boundaries. As previously discussed, in the case of binary classification problems, the linear function is enlarged by integrating it with the logistic function, resulting in the formulation of the overarching function (Horvat et al. 2020).

$$y(X) = W^T \cdot X + w_0$$

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$y(X) = f(\theta^T \cdot X + w_0) = f(z)$$

The parameter z can be defined as

$$z = \theta^T \cdot X + w_0$$

The parameters for the model are w , w_0 , and θ .

3.2.2 Decision Tree

A decision tree is a structure used to express mappings. It is made up of nodes: leaf nodes tagged with a class and test nodes connected to subtrees. Test nodes assess the qualities of instances to decide an outcome, which leads to the relevant subtrees. Instances are categorised by traversing the tree from the root up. When it's a test node, the outcomes direct the traversal, whereas the label of a leaf node predicts the class. Decision trees are created by separating instances based on tests, resulting in hierarchical partitioning. In terms of geometry, characteristics define a description space where instances are points (Jain & Kaur 2017).

Decision surfaces, generally hyperplanes for single-attribute tests, are inserted into tests, dividing space into class-associated areas. More complicated checks can be used to decrease instance subdivisions in order to reduce data fragmentation. Subset testing (e.g., A [c1, c2,...]) is a typical strategy for multivalued nominal characteristics. Multiple attributes, logical combinations, condition sets, or numeric attribute linear combinations (allowing for arbitrary hyperplanes) are used in advanced approaches. Any test option aligns with instances as long as similar attribute-value vectors are not designated differently. Smaller trees, on the other hand, are recommended due to their simplicity of comprehension and potential for improved prediction accuracy. Class distribution analysis and heuristics such as the Gini index or information gain ratio are used to assess test efficacy (Quinlan 1996).

3.2.3 Random Forest

Random forest, a well-known machine learning approach for predictive modelling, works by creating many classification and regression trees using randomised selections of training data and predictor variables. These unique tree predictions are combined to provide forecasts for each occurrence, demonstrating greater predictive accuracy as compared to single decision trees, especially for complex datasets.

When dealing with datasets packed with predictor factors, random forest performs admirably. However, maximising efficiency frequently entails minimising predictor variables. While random forests shine with a wide range of variables, reducing the number of predictors makes for a more effective method. Variable selection, which identifies significant predictors for a more concentrated model, is a critical step in current prediction modelling. This is especially important given that modern datasets usually contain a plethora of possible predictors (Speiser et al. 2019).

3.2.4 K Nearest Neighbors

The standard k-nearest-neighbour (kNN) technique usually assumes a homogeneous distribution of training data across various classes. Nonetheless, in actual applications, real-world events frequently contain unbalanced datasets. An unbalanced dataset contains a dominant majority class that accounts for a large proportion of all cases, whereas the opposing minority class accounts for a tiny proportion. In classification systems based on the kNN approach, the parameter k is crucial. During classification, the k closest neighbours of a test instance from the training set are initially identified. Predictions are then made based on the class labels of these nearest neighbours. The distribution of examples across various classes in the training set is often unequal, with some classes having more instances than others. As a result, the choice of parameter k is quite sensitive, since it has a large influence on classification performance. Using a fixed k value may induce bias, especially for bigger classes (Sun & Huang 2010).

The k-nearest-neighbor (KNN) approach is widely used in classification jobs. It uses a distance measure to determine the dissimilarity or likeness of two occurrences. The most generally used distance measure is the Euclidean distance, abbreviated as $d(x, y)$, which measures the separation between instances x and y as follows: (Jiang et al.

2007).

$$d(x, y) = \sqrt{\sum_{i=1}^n (a_i(x) - a_i(y))^2}$$

3.3 Data Set Description

3.3.1 Banzuke

The notion of "Banzuke" is crucial to the complex ranking system used in sumo wrestling. This method, which is announced before each "basho," or tournament, is a critical predictor of a sumo wrestler's ranking within their separate leagues. The sumo hierarchy is divided into six sections: Makuuchi, Juryo, Makushita, Sandanme, Jonidan, and Jonokuchi. These categories provide a formal framework for categorising wrestlers based on their performance, experience, and achievements. Makuuchi, the highest level, is further broken into numerous categories, including Maegashira, Komusubi, Sekiwake, Ozeki, and Yokozuna. Each ranking is represented by a different acronym, like "Y" for Yokozuna, "O" for Ozeki, and "M" for Maegashira. Furthermore, the letters "e" and "w" represent east and west, with "e" being the better position. This elaborate system concludes with the prestigious title of Yokozuna, whereas Jonokuchi signifies the lowest rank.

The banzuke dataset contains a wide range of variables that give insight into the lifestyles and qualities of sumo wrestlers. Among these aspects is the key "ranking" factor, which represents a wrestler's current status. This rating changes based on a wrestler's performance, with more victories than losses leading to upward mobility and the possibility of promotion to higher categories in subsequent tournaments. Another essential aspect is the "Name" category, which represents the wrestler's ring name. Importantly, this dataset enables wrestlers to change names and even allows numerous wrestlers to share a single ring name. Furthermore, the wrestlers' roots are represented through the "Hometown" feature, which reveals their place of birth. The notion of "Stable Name," a fundamental component of sumo culture, exposes wrestlers' associations with specific training stables, where they dwell and endure rigorous training routines overseen by renowned mentors who are frequently retired wrestlers themselves.

The dataset digs further into personal variables, such as "Date of Birth," providing insights into wrestlers' ages, which range from 17 to 55 years, with an average age of 32. Furthermore, the wrestlers' "height" and "weight" are measured in centimetres and kilogrammes, respectively, representing the physical variety of the sport. The ranges and averages of these measurements—height ranging from 162 cm to 203 cm and weight ranging from 60 kg to 273 kg—add richness to the intricate tapestry of sumo wrestling's different competitors.

3.3.2 Torikumi

The notion of "Torikumi," which defines the highly scheduled timetable of sumo bouts or contests, is central to the thrilling world of sumo wrestling. These enthralling battles conform to a system of strict regulations that are rigorously defined, offering the sport a profound sense of heritage and discipline. The torikumi, as the basis upon which the tournament is built, is a representation of the sport's veneration for its rich history. Torikumi's dynamic character is one of its distinguishing features—the pairings for these enthralling confrontations are revealed just two days before the matches begin, heightening suspense and intrigue. Each tournament, which lasts 15 days and is called as a "basho," is distinguished by a complex combination of strategy, skill, and pure physical power.

The Torikumi dataset is a library of insights into the complex world of sumo wrestling. Within this dataset, a tapestry of information comes to life, with each thread contributing to the colourful fabric of the sport. Among the notable features is the "Player1 Ranking," which provides insight into the rank of the first wrestler, signifying their talent, experience, and performance history. This rank, which is unique to each wrestler, offers a striking image of their position inside the sumo hierarchy. This rating is accompanied by the "Player1 Name," which adds a personal touch by displaying the name of the first wrestler involved in the match. The "Match Details" section sheds information on this wrestler's victories and defeats, enhancing the story of their tournament journey.

The attention paid to the strategies used by sumo wrestlers to attain success is an enthralling aspect of the Torikumi dataset. The "Technique" section reveals the

strategic movements and talents used by the first wrestler, Player 1, to defeat their opponent, Layer 2. The artistry involved in these approaches adds a dimension of complexity to the sport, emphasising the marriage of physical power and tactical acumen. Player1's insights are supplemented with characteristics focusing on Player2, particularly the "Player2 Ranking" and the "Player2 Name". These characteristics mirror the structure observed with Player 1, providing a comprehensive perspective of both players in the fight. Overall, the Torikumi dataset is a treasure trove of data that depicts the spirit of sumo wrestling, including its rankings, rivalries, strategies, and the enthralling choreography of the fights that define this ancient art.

3.4 Evaluation Methods and Measures

In addition to evaluating the performance of data mining algorithms using these two testing methods, it is important to note that the choice of k in the 10-fold CV and the 80% training data split in the percentage split method can vary depending on the specific characteristics of the dataset and the research objectives. Researchers frequently choose these factors with care to achieve a thorough study.

Furthermore, using accuracy as the primary assessment parameter offers a clear indication of how well the algorithms perform in identifying situations. It is worth noting that accuracy is a fundamental metric, but in some cases, other metrics such as precision, recall, or F1-score may also be considered to gain a more nuanced understanding of the algorithm's performance, particularly when dealing with imbalanced datasets or specific application requirements.

Finally, after compiling the outcomes of these assessments, a complete analysis is performed to find the best tool for categorization purposes. This option is critical since it leads the selection of algorithms and methodologies for future applications, guaranteeing effective and accurate data classification (Wahbeh et al. 2011)

In this study, we're trying to predict the outcome of sports games, specifically focusing on the performance of the home team. To measure how well our prediction methods work, we use a metric called accuracy. Accuracy tells us how many correct

predictions we made compared to the total number of predictions. We often use something called a confusion matrix to get a clear picture of how well our prediction model is doing. It helps us see where our predictions match up with the actual results. For example, it can show us how many times we correctly predicted a win or a loss and how many times we got it wrong. To calculate accuracy, we can break it down into parts. We look at true positives (when we correctly predict a positive outcome, like a win), true negatives (when we correctly predict a negative outcome, like a loss), false positives (when we incorrectly predict a positive outcome), and false negatives (when we incorrectly predict a negative outcome) (Horvat & Job 2019).

3.4.1 Confusion Matrix

By documenting actual and expected classifications, a confusion matrix serves as a key tool for evaluating the efficacy of categorization systems (Santra & Christy 2012).

Actual/Predicted	Negative	Positive
Negative	a	b
Positive	c	d

Table 3.1: Confusion Matrix

The matrix contains four significant values in the context of a two-class classifier: "a" represents accurate negative predictions, "b" records instances where the classifier incorrectly predicts positivity, "c" records instances where it incorrectly predicts negativity, and "d" captures correct positive predictions. These statistics provide a full assessment of the categorization system's performance, allowing analysts to rate its accuracy and identify areas that may need improvement.



Figure 3.2: Confusion Matrix for Logistic Regression

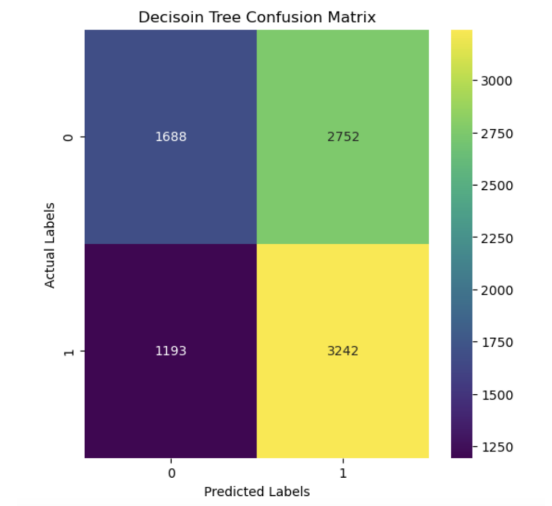


Figure 3.3: Confusion Matrix for Decision Tree

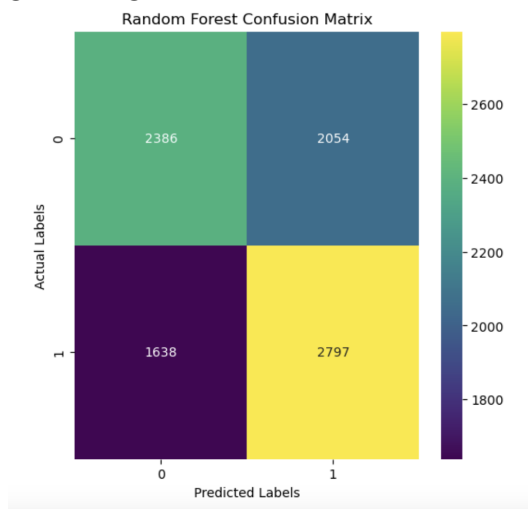


Figure 3.4: Confusion Matrix for Random Forest

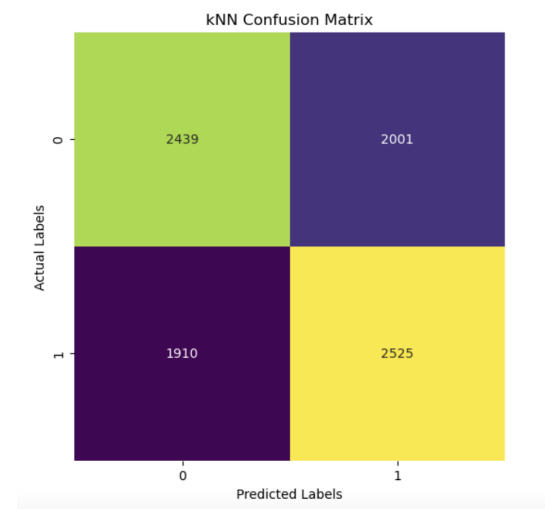


Figure 3.5: Confusion Matrix for kNN

Figure 3.6: Confusion matrix for four classifiers used in the project

3.4.2 Accuracy

In the field of predictive modelling and classification, we come across four key phrases that help us evaluate the effectiveness of our models. The first type of true positive (TP) is when our model precisely predicts a positive result. In the simplest terms, it's a success for the model since it accurately identifies the circumstances in which what we're interested in occurs. On the other hand, we have false positives (FP).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.4.1)$$

This is the polar opposite of a genuine positive. In these circumstances, our model incorrectly predicts a good outcome when it does not occur. It's similar to a false alarm, in which the model incorrectly predicts an event when none exists. The true negative (TN) follows. In these cases, our model properly predicts a bad consequence. It signifies that the model correctly distinguishes instances in which nothing major or noteworthy occurs. The model has made an accurate "no event" call. The last kind is the false negative (FN). This category illustrates our model's mistakes when it predicts a bad result incorrectly. In such circumstances, the model misses anything significant or of interest when it should have. It's the same as losing out on a crucial occurrence because of an error in the model's predictions. These four words are critical in assessing and comprehending the strengths and limitations of prediction models because they reveal how effectively the model works in detecting positive and negative events (Horvat & Job 2019).

3.5 Tools and Resources

Python is the programming language used for this project. Python's adaptability stems from its libraries, which enable for smooth data access, analysis, and change, which is required for continuous data processing in machine learning. Python code is easy to read and comprehend because it is simple. Because of its broad skills in these disciplines, Python is widely used in the development of applications for artificial

intelligence and machine learning.

Jupyter Notebook develops as a vital server-client application that enables code modification and web-based execution within a user-friendly interface. Its adaptability extends to the creation and exchange of files including live code, equations, and dynamic visualisations. Jupyter Notebook's accessibility varies from local use without an internet connection to deployment on distant servers for online access, providing flexibility to both academics and developers.

Pandas takes the lead in data analysis. Its capabilities include data merging, filtering, and aggregation from many sources. Pandas, the leading open-source Python library, is the foundation for data science, analysis, and machine learning projects. It is noteworthy that it is based on Numpy, another programme that supports multi-dimensional arrays, allowing for rapid data manipulation and processing.

Numpy is critical for efficiently storing and managing large amounts of dense data. Its array-based structure effectively stores information, and Numpy facilitates specialised scientific operations such as linear algebra. This package provides significant advantages for computational work due to its memory efficiency and speedier execution when compared to similar data structures in Python.

Matplotlib is used to enhance data presentation and visualisation. Its skills include constructing histograms, graphs, and plots, which are an extension of Numpy's numerical power. Matplotlib emerges as a cross-platform Python library specialised to graphical data representation, boosting the interpretability of complicated datasets.

Seaborn enhances data visualisation by providing a comprehensive toolbox for creating aesthetically appealing and meaningful statistical visualisations. It provides a variety of default styles and colour schemes to improve the aesthetics of statistics charts. Notably, Seaborn is strongly linked with Pandas and builds on the Matplotlib toolkit's roots.

Scikit-learn, a powerful toolkit for basic machine learning tasks including as clustering, regression, and classification, broadens the project's reach even more. Scikit-learn uses the Numpy and SciPy libraries to provide a variety of classification, regression, and clustering algorithms, including support vector machines and random forests, as well

as gradient boosting, k-means, and DBSCAN. The architecture of this library is consistent with the larger Python ecosystem, prioritising compatibility and collaboration across frameworks.

3.6 Summary

In this portion of the article, we carefully investigate a variety of different machine learning approaches that were used in the project. In addition, we investigated the general structure and design of the suggested framework, as well as offering a thorough look at the dataset used in the research. Furthermore, we had a detailed discussion regarding the techniques and resources that were used to carry out this study. Within this context, we undertook an in-depth examination of several approaches that were to be used throughout the study process. Furthermore, we thoroughly studied and discussed the precise metrics that were chosen to assess the performance of the presented models.

Chapter 4

Implementation of Methodology

This chapter explains the implementation specifics of the technique stated in the preceding chapter. This contains information about the data pre-processing, feature engineering, and machine learning classifiers employed.

4.1 Data Preparation and Preprocessing

As previously stated, we have two types of data: *banzuke*, which are player ratings in various tournaments over multiple years, and *torikumi*, which is a list of match results between a pair of sumo wrestlers. Because wrestler-specific characteristics like age, height, weight, ranking, and so on are in the *banzuke* files and match turnout details are in the *torikumi* files, these two sets of files must be merged. We have a dataset spanning the years 1940 through 2023, as well as player rankings and matchup information for all six divisions of sumo wrestling. In this thesis, we do not use all of the datasets. Rather, we pick the dataset based on the following criteria:

1. Selecting data for the last 20 years only; i.e. from year 2002 until 2022 as well as a few months of 2023.
2. Selecting data for wrestlers only from the top division (*Makuuchi*); That means the corresponding division codes are Y, O, S, K and M, which all belong to the *Makuuchi* division.
3. Selecting match details from *Torikumi* only if the players are present in the *Banzuke* files for the corresponding year and venue.

Following that, a number of feature-engineering steps are used, which are explained more below.

4.2 Feature Engineering

The features we include in the final dataset can have a significant impact on the effectiveness of the machine learning classifiers we utilise. Over-feature engineering and the selection of small features may harm performance. We drop information about the player, such as their name, stable name, and place of origin. A stable name and place of origin may include some information, but they are unlikely to effect a player's success. Nonetheless, we chose not to include the wrestler's stable name or location since the ordinal values these qualities may take are rather large, and machine learning algorithms might not be able to extract any useful information from them.

Ordinal Encoding of Division

As discussed, only the top division is considered in this thesis. The following ordinal encoding is applied to the various codes in this division.

- 'Y' (yokozuna): 1
- 'O' (ozeki): 2
- 'S' (sekiwake): 3
- 'K' (komusubi): 4
- 'M' (Maegashira): 5

For example, a rank like "M12w" is split into three parts (three separate columns); Division (M: 5), rank within division (12) and the side (w: 0). This is done using a regular expression matching in python.

Note that the side is also ordinal encoded; west (w) is assigned a value of 0 and east (e) is assigned a value of 1.

Age of Wrestlers from DOB

We know the wrestlers' dates of birth (DOB). It is difficult to implement a date feature as part of a machine learning method. As a result, it is preferable to change this to a numeric age characteristic. Simply subtract the date of birth from the current date to get the number of days they've lived thus far. Finally, dividing this amount by 365 gives the estimated age of each wrestler. This number is a floating point feature, yet it can correctly reflect the age difference between any two wrestlers.

4.3 Final Dataset Preparation

Because we aim to use binary classification to predict who will win a sumo battle between any given pair of wrestlers, each row of the dataset must include information about both athletes as well as who won that related bout. This is obtained by combining data from the banzuke and torikumi databases. This process is explained by the following algorithm:

Algorithm 1 Merge Wrestler Features Algorithm

```

1: for year from 2002 to 2023 do
2:   for all venue in {'hatsu', 'haru', 'natsu', 'nagoya', 'aki', 'kyushu'} do
3:     for all torikumi_file in TorikumiFiles(year, venue) do
4:       for all match in Matches(torikumi_file) do
5:         if AreBothPlayersInBanzuke(year, venue, match) then
6:           merged_data  $\leftarrow$  MergeWrestlerFeatures(match)
7:           AddToMergedDataset(merged_data)
8:         end if
9:       end for
10:    end for
11:  end for
12: end for

```

4.3.1 Data Balancing

A key issue with merging the dataset as described in the technique above is that all players who appear first in the reach row of the torikumi files won the wrestling match, while those who appear second lost the match. As a result, the player who comes first in our final prepared dataset will win. As a result, machine learning classifiers learn to predict the first-arriving player as the winner without taking into account any of their

qualities. This main problem is easily resolved with randomization. That is, when combining the two players, place the winner in the first position with a probability of 0.5 and the losing player in the first position with a probability of 0.5 as well. Add a target label column with a value of 1 if the first player won the bout and 0 otherwise. This ensures an approximately 50-50 data mix between the two groups. Using a binary classifier now makes sense.

4.3.2 Data Normalization

When we examine the dataset thus far, we can see that certain characteristics, such as weight and height, accept high values, while others, such as side, take tiny values (values 0 or 1). As a result, it is often recommended to normalise the dataset since some methods, such as logistic regression, are sensitive to changes in feature weights. There are several types of data normalisation systems, such as min-max normalisation, z-score normalisation, and so on. We apply a basic max normalisation, which divides each column by its maximum value. As a result, the highest value in any column after normalisation (also known as scaling) will be 1. Negative values are permitted in this normalisation technique, but they must be between 0 and -1.

4.4 Train-Test Data Split vs Cross Validation

The final data set is separated into a train dataset and a test dataset in an 80% to 20% ratio. The machine learning models are trained exclusively on the training dataset, and the evaluation results are presented on both the training and test datasets.

Cross-validation is another method for assessing the performance of machine learning classifiers. We used 10-fold cross-validation. This stage employs the whole dataset, and the average results across all ten folds are presented. To choose the optimal hyperparameters, we also used cross-validation and performed hyper-parameter tweaking, as mentioned below. The selected hyperparameters are utilised in the train-test split dataset mentioned above to report the best performance.

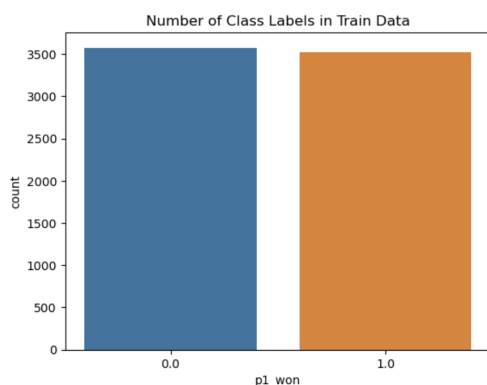


Figure 4.1: No. of class labels in train data (80% split)

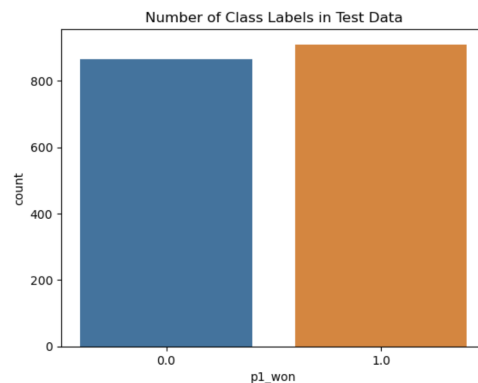


Figure 4.2: No. of class labels in test data (20% split)

4.5 Hyper-parameter Tuning

Each machine learning algorithm has its own set of hyperparameters. Choosing the best hyperparameter for the test dataset can result in optimal performance. The value of k (number of neighbours) in the kNN classifier and the "max-depth" in a decision tree classifier are examples of hyper-parameters.

To execute hyper-parameter tweaking, we may run a search over various hyperparameter options and select the one that provides the greatest cross-validation performance.

4.6 Training Machine Learning Classifiers

The scikit-learn package is used to train and evaluate machine learning classifiers. Each classifier's "fit" function trains the classifier for a given set of training features and training labels. The testing can begin by guessing the class labels with the "predict" function.

The plots are created in Python using Matplotlib and the Seaborn packages.

4.7 Summary

This chapter outlines all of the procedures involved in developing a machine learning model for predicting sumo fight winners. Cleaning the dataset, feature engineering, and training machine learning classifiers are all part of this process.

Chapter 5

Evaluation Result and Discussion

This chapter describes the evaluation and results of the different methods described in this thesis. A comparison of each classifiers is provided. The parameter settings for the classifiers are also detailed so that results can be reproduced.

5.1 Logistic Regression

The 'LogisticRegression' model from the sklearn package is used. A maximum iteration of 500 was used. All other parameters are set to be the default parameters, which means that the solver or optimizer used is the 'lbfgs', the tolerance value for convergence of optimization is $1e-4$, l2 penalty is used for regularization and the regularization constant used is $C = 1.0$.

The accuracy on the train set is 59.11% and the accuracy on the test set is 59.72%.

5.1.1 10-fold Cross Validation

The dataset was divided into ten sets for 10-fold cross validation. The sklearn package's 'crossvalidate' function is used to conduct k-fold cross validation. The mean average accuracy, precision, recall, and f1-score are calculated.

- Accuracy: 0.589
- Precision: 0.593
- Recall: 0.609
- F1-score: 0.6

5.2 Decision Tree

The DecisionTreeClassifier from sklearn package is used to train a decision tree classifier. All default parameters are used except for the max-depth parameter which is set to be 2 by default. But this hyper-parameter is tuned to obtain best results later.

5.2.1 K-fold Cross Validation

The mean average accuracy, precision, and recall and f1-score are

- Accuracy: 0.536
- Precision: 0.548
- Recall: 0.484
- F1-score: 0.513

5.2.2 Hyper-parameter Tuning

The different values taken by the hyper-parameter 'max-depth' are 2, 4, 6, 8, 10, 12 and 14. Hyper-parameter tuning selected the value that gave the highest f1-score value. The chosen max-depth value is 2.

5.3 K Nearest Neighbors Classifier (kNN)

The KNeighborsClassifier from the sklearn package is used. The only hyper-parameter tuned is the number of neighbors (k).

5.3.1 K-fold Cross Validation

The results from cross validation are shown below.

- Accuracy: 0.537
- Precision: 0.543
- Recall: 0.544
- F1-score: 0.544

5.3.2 Hyper-parameter Tuning

The number of neighbors being tuned ranged from 3, 5, 7 upto 15. The best parameter value obtained is $k = 13$.

5.4 Random Forest Classifier

A random forest classifier is trained and tested using the RandomForestClassifier from the sklearn package. The train and test set accuracies obtained are 55.49% and 55.49% respectively.

5.4.1 K-fold Cross Validation

The steps of cross validation are the same as described earlier. The results from cross validation are shown below.

- Accuracy: 0.565
- Precision: 0.561
- Recall: 0.672
- F1-score: 0.607

5.4.2 Hyperparameter Tuning

The different values taken by the hyper-parameter 'max-depth' are 2, 4, 6, 8, 10, 12 and 14. Hyper-parameter tuning selected the value that gave the highest f1-score value. The chosen max-depth value is 2.

5.5 Results: K-fold Cross Validation

Various machine learning techniques, including logistic regression, k closest neighbours, decision trees, and random forest classifiers, were subjected to 10-fold cross-validation. For this reason, the dataset was separated into ten sets, and the average of the ten folds is presented. Table 5.1 summarises the findings of the K-fold cross validation, including accuracy, precision, and recall.

ML Algorithm	Accuracy (%)	Precision	Recall	F1-score
Baseline Classifier	56.39	-	-	-
Logistic Regression	58.9	59.3	60.9	60
K Nearest Neighbors	53.7	54.3	54.4	54.4
Decision Tree	53.6	54.8	48.4	51.3
Random Forest	57.3	57.5	62.2	59.3

Table 5.1: K-fold Cross Validation Results

The table summarises each algorithm's performance in terms of accuracy, precision, recall, and f1 score measures. Among the tested methods, logistic regression had the best accuracy, while decision tree had the lowest.

5.5.1 Comparison of Classifiers

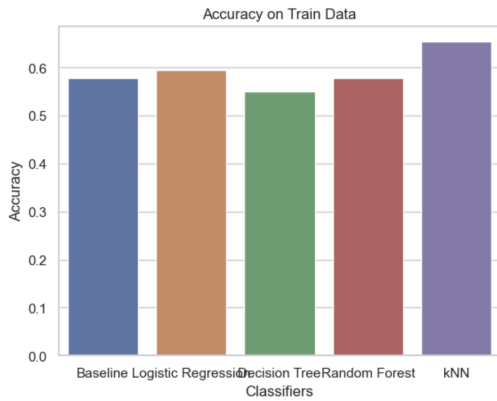


Figure 5.1: Accuracy of Train Data



Figure 5.2: Accuracy of Test Data

A pretty unusual and noteworthy phenomenon has developed in the framework of our research: test data accuracy frequently outperforms training data accuracy, as seen in the two figures 5.1 and 5.2 supplied. This surprising and paradoxical finding has far-reaching consequences. Typically, one would expect the model to thrive on the data it was expressly trained on, with any deviations raising worries about over fitting or the model's capacity to generalise. The observed reversal in accuracy trends, on the other hand, provides a variety of intriguing possibilities. To begin with, it suggests that our model captures a highly generalised representation of the underlying data patterns. This implies that our model does more than just memorise the training data; it also derives essential properties that apply reliably to unseen test data, indicating its potential for real-world application. Second, it might imply that the model's performance has been impacted by the training data, which looks to be more difficult or

noisy. When evaluated on simpler, cleaner test data, the model may have met a more diversified set of cases during training, resulting in an apparently lower accuracy. As a result, we must examine the quality and representativeness of our training dataset. Finally, this event may highlight the efficacy of the regularisation strategies used during model training. These strategies appear to improve the model’s ability to generalise to new, previously unknown data, but at the expense of a minor decrease in accuracy on the training set. In light of these intriguing findings, our research aims to examine and deconstruct the reasons influencing this accuracy reversal. We want to learn more about the model’s resilience, the quality of our training data, and the influence of different regularisation approaches in order to assure the model’s dependability and applicability for real-world applications.

5.5.2 Conclusions

The performance of several machine learning methods was examined in this part using K-fold cross-validation and hyper-parameter tweaking. The findings shed light on the accuracy and possibility for improvement of each method. Logistic regression was shown to be the most accurate model, while other algorithms such as K closest neighbours, decision trees, and random forests also performed well.

The use of hyper-parameter tuning shows that little changes to model parameters can result in increased accuracy. However, the impact differed amongst algorithms, with some seeing greater increases than others.

Overall, this study establishes the foundations for future optimisations and advancements by providing a platform for selecting appropriate machine learning algorithms for specific use cases.

Chapter 6

Conclusion

In conclusion, the use of machine learning techniques in this study marks a significant advancement in the field of sumo wrestling outcome prediction. We have showed the capacity to untangle the complexity of this ancient sport and shed light on the elements impacting winning by utilising a varied variety of techniques, including logistic regression, k-nearest neighbours, decision trees, and random forests.

Our rigorous approach to model assessment, which includes both train-test splits and k-fold cross-validation, assures the predictability and generalisation capabilities of our prediction models. This not only strengthens the legitimacy of our findings, but it also lays the groundwork for future study in this area.

The quest for better accuracy through hyperparameter tweaking demonstrates our dedication to pushing the boundaries of prediction performance. While the current accuracy ratings for logistic regression and random forests are 58.9 percent and 57.3 percent, respectively, there is still opportunity for improvement. As we explore further into the dynamics of sumo wrestling, the introduction of new characteristics, particularly those relating to the unique strategies used by winning rikishi, shows promise for improving forecast accuracy. One promising route for future studies is to expand our dataset to include lower-division ranks. This larger view might provide useful insights on the evolution of sumo wrestling methods and the elements that lead to success at various levels of competition. By incorporating data from all divisions, we may discover subtleties and patterns that were previously missed, possibly leading to more accurate and robust prediction models.

In essence, this study marks a first step towards comprehending the complicated world of sumo wrestling via the lens of machine learning. We believe that as we develop

our models, extend our dataset, and study the complex tapestry of sumo tactics, our prediction powers will only grow, providing a great resource for sumo lovers, analysts, and stakeholders. This endeavour demonstrates the power of data-driven techniques in unravelling the secrets of even the most traditional sciences.

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